EVIDENCE-BASED | VOLUME SCIENCE | 3 PROGRAMS | PERIODIZATION

THE SCIENCE OF MUSCLE HYPERTROPHY COMPLETE TRAINING MANUAL

Mechanisms · Volume Landmarks · Progressive Overload · Periodization · 3 Full Programs

Evidence-based training science for maximum lean muscle growth

ELITE FITNESS GUIDE SERIES | Research-Based • Professional Edition • 2024

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■ 1. THREE MECHANISMS OF MUSCLE HYPERTROPHY

Muscle hypertrophy (growth) results from three distinct but overlapping stimuli, identified and refined by Dr. Brad Schoenfeld's seminal 2010 review in the Journal of Strength and Conditioning Research:

Mechanism	Definition	How to Train for It	Evidence Level
Mechanical Tension	Force generated during muscle contraction and stretch; the primary driver of hypertrophy	Heavy loads (65–85% 1RM); full range of motion; controlled eccentric phase; compound movements	Strongest evidence; most important driver
Metabolic Stress	Accumulation of metabolites (lactate, hydrogen ions, phosphate) in muscle; 'the pump'	Moderate loads (65–75% 1RM), short rest periods (60–90s), isolation exercises, occlusion/BFR training	Moderate evidence; meaningful contributor
Muscle Damage	Micro-tears in muscle fibers (especially eccentric phase) triggering satellite cell activation and repair/growth	Eccentric emphasis, novel exercises, extreme stretch positions; lower frequency (recovery needed)	Emerging evidence; may contribute but not required

> ■ *Dr. Brad Schoenfeld's meta-analyses (2017, 2019) demonstrate that volume-equated protocols emphasizing mechanical tension produce equal or superior hypertrophy to high-metabolic-stress protocols, confirming mechanical tension as the primary driver.*

■ 2. VOLUME LANDMARKS — MEV, MV, MAV, MRV

Volume landmarks describe the range of weekly sets per muscle group that produce specific training responses. These concepts, developed by Dr. Mike Israetel (Renaissance Periodization), provide a framework for programming volume rationally rather than arbitrarily.

Landmark	Acronym	Definition	Typical Range (sets/muscle/week)	Application
Minimum Effective Volume	MEV	Minimum sets needed to make gains above maintenance	4–8 sets	Deload weeks; returning from break; beginners
Maintenance Volume	MV	Sets needed to maintain current muscle mass with no effort to grow	6–10 sets	Cutting phases; competition prep; high-stress periods
Maximum Adaptive Volume	MAV	The 'sweet spot' — volume range producing the most hypertrophy	10–20 sets	Primary training volume for most cycles
Maximum Recoverable Volume	MRV	Maximum sets from which you can recover before the next session	15–25+ sets	Upper limit; approach only in peak weeks; AAS elevates this

Practical Application:

- Start a new training block at MEV; add 2 sets/muscle/week each week
- When performance begins declining (DOMS excessive, strength drops, sleep worsens) — you've exceeded MRV
- After 4–6 weeks at MAV, deload back to MEV; reset and begin again with slightly higher MAV
- Natural athletes: MRV ≈ 15–25 sets/muscle/week. On AAS/SARMs: MRV increases to 25–35+ sets

■ 3. PROGRESSIVE OVERLOAD — THE MASTER VARIABLE

Progressive overload is the single most important variable in hypertrophy training. Your body only adapts (grows) when subjected to a stimulus **greater than previous stimuli**. Without overload, you maintain — never grow.

Method	Description	Best Applied To	Rate of Progress
Load Progression	Add weight to the bar each session or week	Compound movements (squat, bench, deadlift, OHP)	2.5–5 kg/session for beginners; monthly for advanced
Rep Progression	Add reps within target rep range before increasing weight	All exercises; when load jumps are too large	1–2 extra reps per session; increase load when top of range hit
Set Progression	Add a working set each week (volume accumulation)	When load/rep progress stalls	1 set/exercise/week added over a mesocycle
Density Progression	Same work in less time (reduce rest periods)	Metabolic training; body composition phase	5–10 seconds less rest per session
ROM Progression	Increase range of motion over time	Mobility-limited movements; deficit deadlifts	Small incremental ROM increase per session
Technique Progression	Better execution = more stimulus from same load	All exercises (especially beginners)	Continuous; reduces injury risk while increasing effective load

■ 4–6. REP RANGES · FREQUENCY · REST PERIODS

Rep Range	% 1RM	Primary Adaptation	Best Exercises
1–5 reps	90–100%	Neural strength; maximal force production; some hypertrophy	Powerlifts (squat, bench, deadlift, OHP); cleans
6–10 reps	75–85%	PRIMARY hypertrophy zone; strength-hypertrophy blend	Compound movements + heavy isolation (DB curl, leg press)
10–15 reps	65–75%	Hypertrophy-focused; metabolic stress; excellent for isolation	All isolation movements; cables; machines
15–30 reps	50–65%	Metabolic stress; endurance; useful for volume work	Cables, bands, light isolation; finisher sets

■ *Schoenfeld 2017 meta-analysis: hypertrophy is statistically equal across rep ranges 1–30 when volume is equated (sets × reps × weight). However, higher loads (6–12 reps) achieve the same hypertrophy in fewer sets, making them more time-efficient.*

Training Frequency: Each muscle group responds to protein synthesis stimulation approximately **every 48–72 hours**. Training a muscle 2× per week allows 2 protein synthesis spikes vs once per week = theoretically 2× the growth stimulus. For most natural athletes: 2× per week per muscle group is optimal. 3× is marginal benefit for advanced athletes.

Rest Periods: 3–5 minutes for compound/strength sets (allows full ATP-PCr recovery); 1–2 minutes for isolation/metabolic sets (maintained metabolic stress). Shorter rest ≠ better hypertrophy when using free weights — full recovery enables heavier sets = more tension.

■ 7. PERIODIZATION MODELS

Model	Structure	Best For	Frequency of Change
Linear Periodization	Volume decreases as intensity increases over weeks/months	Beginners and powerlifters; predictable strength progression	Every 3–6 months (full cycle)
Undulating Periodization (DUP)	Volume and intensity vary weekly or daily	Intermediate-advanced; prevents neural adaptation plateau	Weekly or daily
Block Periodization	Distinct training blocks (accumulation → intensification → realization)	Advanced athletes; competition preparation	Every 3–6 week blocks
Autoregulation (RPE-based)	Load and volume adjusted daily based on readiness (RPE scale)	Advanced athletes; variable life stress; most sophisticated	Daily

■ 8–9. MIND-MUSCLE CONNECTION & DELOADS

- **Mind-muscle connection (MMC):** Deliberately focusing attention on the target muscle during contraction increases muscle activation by 15–20% (Snyder & Leech, 2009). For isolation exercises, MMC is critical — for compound movements, external focus ('push the floor away') produces more force
- **Stretch-mediated hypertrophy:** New research (2022–2024) demonstrates that exercises producing maximum tension at full stretch (spider curls, incline curls, Romanian deadlifts, deep squats) produce superior hypertrophy vs exercises only loading the shortened position
- **Eccentric emphasis:** Eccentric phase (lowering) produces greater muscle tension per motor unit recruited — slowing the eccentric to 2–4 seconds increases hypertrophic stimulus significantly

Deload Protocol: After 4–6 weeks of hard training, perform a 1-week deload: reduce volume by 50% (keep same exercises and intensity), reduce sets to 2–3 per exercise, eliminate drop sets and intensification techniques. Deloads prevent cumulative fatigue from masking fitness gains and allow connective tissue recovery.

■ 10. PROGRAM 1 — PPL 5-DAY SPLIT

Day	Focus	Exercises	Sets × Reps
Monday	Push (Chest/Shoulders/Triceps)	Barbell Bench Press · Incline DB Press · Cable Lateral Raises · Shoulder Press · Tricep Pushdown · Skull Crushers	4×6–8 · 3×10–12 · 3×15–20 · 3×10–12 · 3×12–15 · 3×12–15
Tuesday	Pull (Back/Biceps)	Barbell Row · Pull-Ups/Pulldowns · Seated Cable Row · Face Pulls · EZ-Bar Curl · Hammer Curls	4×6–8 · 4×8–12 · 3×10–12 · 3×15–20 · 3×10–12 · 3×12–15
Wednesday	Legs (Quads/Hams/Calves)	Barbell Squat · Leg Press · Romanian Deadlift · Leg Curl · Walking Lunges · Calf Raises	4×6–8 · 4×10–12 · 4×10–12 · 3×12–15 · 3×12–15 · 5×15–20
Thursday	Push 2 (Shoulders/Chest/Triceps)	OHP/Arnold Press · Cable Flyes · Incline Bench · Overhead Tricep Extension · Lateral Raises	4×8–10 · 3×10–12 · 3×12–15 · 3×10–12 · 3×12–15 · 4×15–2
Friday	Pull 2 (Back/Biceps/Rear Delts)	Deadlift · Wide Pulldown · Single-Arm DB Row · Reverse Flyes · Preacher Curl · Incline DB Curl	4×4–6 · 4×8–12 · 3×10–12 · 3×15–20 · 3×10–12 · 3×12–15
Sat/Sun	Rest + Optional Cardio	30–45 min LISS walking/cycling	N/A

■ 11–12. UPPER/LOWER 4-DAY & FULL BODY 3-DAY

Day	Focus	Key Exercises	Sets × Reps
Monday	Upper — Strength	Bench Press · OHP · Barbell Row · Pull-Ups · Bicep Curl · Tricep Extension	4×4–6 · 4×4–6 · 4×4–6 · 4×6–8 · 3×8–10 · 3×8–10
Tuesday	Lower — Strength	Squat · Romanian Deadlift · Leg Press · Leg Curl · Calf Raises	4×4–6 · 4×6–8 · 4×8–10 · 3×8–10 · 4×15–20
Thursday	Upper — Hypertrophy	Incline DB Press · Lateral Raises · Seated Row · Pullovers · Preacher Curl · Pushdowns	3×10–15 each — focus on contraction quality and MMC
Friday	Lower — Hypertrophy	Front Squat/Hack Squat · Bulgarian SSQ · Leg Ext · Lying Leg Curl · Seated Calf	3×12–15 each — full ROM; slow eccentric

Day	Exercises	Sets × Reps
Monday (Full Body A)	Squat 3×5 · Bench 3×5 · Row 3×5 · OHP 2×10 · Curl 2×12 · Push-up 2×max	Progressive overload on main 3 lifts each session
Wednesday (Full Body B)	Deadlift 1×5 · OHP 3×5 · Chin-Up 3×5 · DB Bench 2×10 · Face Pull 2×15 · Plank 2×60s	Alternate A/B; add 2.5kg each session when all reps completed
Friday (Full Body C)	Front Squat 3×8 · Incline Press 3×8 · Pendlay Row 3×8 · Dips 2×10 · Reverse Curl 2×12	Technique focus day — videos or mirror

Progressive overload without periodization leads to plateau. Periodize your training — systematically vary volume and intensity every 4–6 weeks for continuous progress.